

Supporting Information for “A grouped calibration test for logistic regression that tolerates a few corrupted records”

Ebrahim Khaled Ebrahim, Osama Abd El-Aziz Hussein and Ahmed El-Kotory

S0 The comparisons specified before the study was run

Before any scenario was computed, seven comparisons were written down with their scenarios, their statistic, a threshold and a rule for what would count against EDGE, and the document was fixed by its SHA-256 hash. Table S1 lists them with their outcomes. Three went against the test: H1 and H3, where the GiViTI belt is the better test when events are scarce or the curve bends once, and H4, where EDGE at ten groups led or tied the best of four partition tests in 17 of the 19 selected scenarios, where all 19 were required; at the default partition it did so in all 21. The main text reports each of them as a finding in Section 5.2 and as a usage boundary in Section 8.3. The hash-fixed documents are in the reproducibility archive.

The main study re-ran, under these documents, a battery that had been run before them. The comparison with robust tests (Section 6.6), the external-validation, streaming and what-protects studies were added after the main results had been read, each fixed by its own hash before any of its replicates ran. Its rule was that the Mallows form of the robust Stukel test would be said to match EDGE’s protection if its false-alarm rate stayed at or below 0.10 with ten corrupted records in 1000 under both errors. It did not: 0.168 and 0.789.

Table S1: The seven hypotheses declared before the study was run, and their outcomes. The statistic and the threshold are those fixed in the pre-declaration; no threshold was moved after the results were seen. The last column counts the scenarios in which EDGE leads or ties, which is a different question from the declared verdict and answers it differently: H1 fails on a condition about one design while the test leads or ties in 26 of 32 scenarios. Section 5.2 of the main text reports the results as findings.

Hypothesis	Statement	Scenarios	Statistic	Threshold	Outcome	Leads or ties
H1	EDGE-poly3 leads GiViTI on symmetric tails	32	0.105	0.1	against	26 of 32
H2	EDGE-sym is not behind Stukel’s joint score	32	0.005	-0.014	holds	25 of 32
H3	EDGE-poly3 is not far behind GiViTI on asymmetric links	21	-0.075	-0.05	against	7 of 21
H4	EDGE-poly3 leads or ties the partition tests at $G = 10$	29	17.000	19	against	17 of 19
H4-ruleG	the same at the default partition	29	21.000	19	reported	21 of 21
H5	EDGE-sym is not behind Stukel’s one-parameter score	4	0.006	0.02	holds	2 of 4
H5-skew	the same under a skewed covariate	4	0.252	–	reported	0 of 4

S1 The choice of weighting

Which risk groups carry the test depends on the weighting, and the right weighting depends on the basis. For the symmetric basis the score weighting is what makes the test work at high discrimination. It leads the unit form in 21 of the 32 power scenarios, and against a probit truth the difference is large: power rises from 0.096 to 0.598 at $n = 3500$, and from 0.506 to 0.834 at $n = 5900$, at nominal size in both forms. These are the two scenarios in which this basis, in its unit form, falls behind Stukel’s joint score test by 0.352 and 0.214. With the score weighting the order reverses and EDGE leads by 0.150 and 0.114.

Scaling by $\sqrt{V_g}$ moves weight towards the extreme risk groups, and the symmetric shape $\eta|\eta|$ is carried by exactly those groups. The cubic basis already reaches them through its own columns, so the same rescaling buys it nothing and costs it variance: for the cubic basis the score form is behind in 24 of the 32 scenarios and ahead in one, by a mean of 0.039.

One regime overrides both rules. The score weighting leans on the extreme groups, and those are the groups a corrupted record occupies: with 25 of 1000 covariates corrupted, the cubic basis in its score form reached a false-alarm rate of 0.894 where its unit form reached 0.344. When the predictors may contain corrupted records, the unit form is required whatever the basis.

S2 The three further corruptions

When the k highest-risk records have their outcome set to zero (C2), every test rises together, EDGE included: at ten such records in a thousand it reads 0.667 against Stukel’s 0.718 and the cubic test’s 0.837. A missed event at the top of the risk scale is the pattern a calibration test is built to find.

A unit error that makes a prediction milder rather than extreme (C3) moves no test under the logistic truth, as the mechanism of Section 3 of the main text predicts, since such a record lands in a crowded middle group rather than a sparse extreme one. Under the two misfit truths it costs some power: sixteen of the 168 test-by-scenario readings move by more than three nominal standard errors, ten under the complementary log–log truth and six under the probit, and EDGE is among them. The largest move is its own, 0.818 falling to 0.770 at ten corrupted records under the complementary log–log truth. These moves are losses of power against a real misfit, not false alarms. Duplicated high-risk records (C4) leave every test between 0.054 and 0.071, and the Hosmer–Lemeshow variants between 0.040 and 0.057.

S3 The resampling tests under corruption

The two resampling tests are too slow for the main contamination design, so a separate study put them on a smaller one: three records in five hundred with their covariate multiplied by four or by eight, 100 replicates a scenario, and every test on the same data sets. With 100 replicates the Monte Carlo standard error of a rate of 0.05 is 0.022, so a rise of a few points cannot be seen. Multiplying by eight carries a corrupted record’s linear predictor to a median of 11 and at most 16.5. Table S2 gives the result.

BAGofT and EDGE with the cubic basis gave no sign of moving: 0.06, 0.01 and 0.05 for the first in the clean, $\times 4$ and $\times 8$ scenarios, and 0.08, 0.02 and 0.05 for the second; with 100 replicates a rise of a few points could not be seen. Le Cessie’s smoothed test reads 0.07, 0.04 and 0.07. The Liu projection test holds at $\times 4$ and breaks at $\times 8$, where it reads 0.22; on the same replicates it raised 18 false alarms that EDGE did not, against one the other way (Holm-corrected McNemar $p < 0.001$). The record-level score and calibration tests sit between 0.29 and 0.49 in both corrupted scenarios. The symmetric basis reads 0.20 at $\times 8$. This is consistent with the main text: the cubic basis, here at the default partition (twenty groups), did not move, while the symmetric basis did at $\times 8$.

Table S2: False-alarm rate at the nominal 5% level when three records of $n = 500$ have their covariate multiplied by four or by eight and the model is correct for every other record. 100 replicates a scenario, every test on the same data sets; the nominal standard error is 0.022. EDGE is in its unit form at the default partition, which is twenty groups here. The Liu projection test and BAGofT reject at $p < 0.05$, the others at $p \leq 0.05$. BAGofT pools the residuals within groups it learns from the data and le Cessie’s test over neighbourhoods in covariate space, so both are listed with the pooled tests; le Cessie’s test was run on the same data sets. The Liu projection test took a median of 17 seconds a data set and BAGofT 371.

Test	clean	$\times 4$	$\times 8$
<i>pooled</i>			
EDGE (default)	0.080	0.020	0.050
EDGE, symmetric basis	0.050	0.060	0.200
Hosmer–Lemeshow, $G = 10$	0.020	0.000	0.060
BAGofT	0.060	0.010	0.050
le Cessie–van Houwelingen	0.070	0.040	0.070
<i>record-level</i>			
Liu projection test	0.080	0.060	0.220
Stukel joint score	0.060	0.450	0.490
Stukel one-parameter	0.050	0.430	0.470
Stukel LR refit	0.070	0.330	0.410
GiViTI belt	0.070	0.290	0.370
cubic calibration LR	0.060	0.350	0.400

S4 The robust fit

Every data set of this study was fitted twice, by maximum likelihood and by the Huber-type quasi-likelihood estimator of Cantoni and Ronchetti (2001) (Huber’s ψ on the Pearson residual at 1.345, no weight on the covariates, **robustbase**’s default), which bounds the influence of the residual but not of the covariates, and every test was applied to each fit. Table S3 gives the result.

Five of the ten tests fail to hold their level after a robust fit on uncontaminated data: Stukel’s two score tests, at 0.199 and 0.223, and three of EDGE’s four forms, the symmetric basis in its unit form (0.164) and both score-weighted forms (0.619 and 0.225). The rule fixed in advance was that a failed level check leaves the study measuring that mismatch rather than the contamination; we depart from it in one restricted way by reading the contaminated columns for the five tests whose level survives. A robust fit does not bring the record-level tests back to the nominal level.

Of those five, none has its false alarms reduced. At ten corrupted records the GiViTI belt reads 0.379 under maximum likelihood and 0.400 under the robust fit, the cubic calibration test 0.465 and 0.466, and EDGE 0.089 and 0.087. Several are worse at twenty-five: the Hosmer–Lemeshow statistic rises from 0.114 to 0.171 and the belt from 0.639 to 0.678. A corrupted record’s prediction is formed at its corrupted covariate, so refitting does not reach it; restoring the slope, if anything, makes that prediction more extreme. Two further arms change nothing: at $n = 5000$ with twenty-five corrupted records EDGE reads 0.144 under both fits, and against a weak probit departure the robust fit does not pay for itself either. That a robust fit does not repair a non-robust test is the observation on which robust testing was founded (Hampel et al., 1986; Heritier and Ronchetti, 1994), and Cantoni and Ronchetti (2001) pair their estimator with robust quasi-deviance tests for that reason.

Table S3: Rejection rate at the nominal 5% level under a correct logistic model with k corrupted covariates, $n = 1000$, when the model is fitted by maximum likelihood (ML) and by the Huber-type quasi-likelihood estimator of Cantoni and Ronchetti (2001) — `robustbase`’s `Mqle` at tuning constant $c = 1.345$, with no weight on the covariates — on the same data. This study was run separately from that of the main contamination table and on its own data sets, so its maximum-likelihood columns differ from the corresponding rows of that table by Monte Carlo error alone. The robust fit converged in all 1000 replicates of every configuration and no replicate was excluded. With $k = 0$ the rates are sizes. The robust fit leaves every test’s false-alarm rate where it was or raises it, and it breaks the level of five of the ten tests the gate covers, even without contamination — Stukel’s two score tests and three of the four forms of EDGE — because their reference is built on the maximum-likelihood information matrix. Only the default unit form of EDGE, the two Hosmer–Lemeshow statistics, the belt and the cubic test are read against a reference that still holds. Eleven tests were run and ten are shown: Stukel’s likelihood-ratio refit is omitted because a robust fit has no deviance, so under the robust arm that refit compares a maximum-likelihood augmented fit against a robust base and is not a likelihood-ratio test. 1000 replicates; the nominal standard error is 0.007.

Test	$k = 0$		$k = 10$		$k = 25$	
	ML	robust	ML	robust	ML	robust
EDGE (default)	0.059	0.059	0.089	0.087	0.345	0.339
EDGE, symmetric basis	0.044	0.164	0.055	0.218	0.105	0.520
EDGE, score-weighted	0.056	0.619	0.659	0.560	0.899	0.737
EDGE, symmetric score-weighted	0.050	0.225	0.077	0.272	0.077	0.543
Hosmer–Lemeshow	0.059	0.050	0.045	0.063	0.114	0.171
HL, Farrington-corrected	0.059	0.051	0.046	0.059	0.116	0.140
GiViTI belt	0.049	0.054	0.379	0.400	0.639	0.678
cubic calibration LR	0.051	0.053	0.465	0.466	0.773	0.774
Stukel joint score	0.064	0.199	0.609	0.701	0.892	0.937
Stukel one-parameter	0.045	0.223	0.588	0.686	0.882	0.925

S5 Partitions that pool the extreme groups

Corrupted covariates produce extreme predictions, so they collect in the two extreme groups. A partition that pools only those groups, and keeps the default partition in the middle where most of the power lies, was studied as an alternative to ten groups: EDGE-FR(c) puts the outer $5c$ of the records at each end of the risk scale into one group. It does not have equal-frequency groups, so its level was checked by simulation; it reads 0.039 and 0.050 on clean data at the two sample sizes, and 0.041 and 0.054 on a fresh set of replicates. Table S4 gives the result. At $c = 1\%$ it matches ten groups up to ten corrupted records in 1000 and fifty in 5000 (0.081 against 0.085, and 0.111 against 0.107), is less protected beyond (0.140 against 0.101 at twenty in 1000, 0.431 against 0.328 at a hundred in 5000), and has the same power: a size-adjusted mean of 0.490 against 0.483 over the six link and tail scenarios at $n = 1000$, where the default partition has 0.500. A wider tail costs power, a mean of 0.449 at $c = 2\%$ and 0.408 at $c = 4\%$, without protection in proportion: at $c = 2\%$ it reads 0.104 with twenty corrupted records in 1000 but 0.338 with a hundred in 5000, the same rate. The main text therefore recommends the simpler ten-group partition.

Table S4: The tail-pooled setting EDGE-FR, at three tail fractions, against the default partition and against ten groups. Its default column repeats the EDGE column of Table S5 for the eight scenarios the two tables share, so that the comparison can be read in place. EDGE-FR(c) puts the outer $5c$ of the risk scale at each end into one group and leaves the middle at the default partition, which is used throughout otherwise. 1000 replicates a scenario, the same data sets for every setting. The false alarms are rejection rates at the nominal 5% level under a correct logistic model with k records carrying a corrupted covariate; the rows at $k \geq 20$ for $n = 1000$ and $k \geq 100$ for $n = 5000$ were run on fresh seeds, whose uncontaminated scenarios read between 0.034 and 0.062 for every setting. The power rows are size-adjusted, each setting at its own critical value read from the matched null scenario, and the first is the mean over six link and tail scenarios at $n = 1000$ (cauchit, log-log, three Stukel designs, t_4); the paired readings behind them are in the archive. Widening the tails costs power without adding protection in proportion, and the protection ends in every setting once the corrupted records pass a few tens.

Scenario	default	EDGE-FR			ten groups
		$c = 1\%$	$c = 2\%$	$c = 4\%$	
<i>false alarms, k records with a corrupted covariate</i>					
$n = 1000, k = 1$	0.056	0.064	0.059	0.056	0.056
$n = 1000, k = 2$	0.059	0.053	0.049	0.048	0.053
$n = 1000, k = 5$	0.057	0.047	0.053	0.056	0.051
$n = 1000, k = 10$	0.097	0.081	0.079	0.050	0.085
$n = 1000, k = 20$	0.227	0.140	0.104	0.100	0.101
$n = 1000, k = 40$	0.714	0.397	0.226	0.169	0.234
$n = 5000, k = 5$	0.050	0.041	0.043	0.052	0.048
$n = 5000, k = 10$	0.060	0.045	0.045	0.055	0.047
$n = 5000, k = 25$	0.151	0.059	0.059	0.056	0.065
$n = 5000, k = 50$	0.740	0.111	0.113	0.097	0.107
$n = 5000, k = 100$	1.000	0.431	0.338	0.246	0.328
$n = 5000, k = 200$	1.000	0.987	0.956	0.863	0.945
<i>level, no corruption</i>					
$n = 1000$	0.045	0.039	0.043	0.050	0.040
$n = 5000$	0.048	0.050	0.041	0.048	0.051
<i>power against a real misfit</i>					
six link and tail scenarios, mean	0.500	0.490	0.449	0.408	0.483
complementary log-log, $n = 1000$	0.826	0.824	0.778	0.662	0.812
probit, $n = 5000$	0.278	0.239	0.196	0.146	0.223

Sharper remedies were considered first. Dropping the outer one or two per cent of records,

Table S5: False-alarm rate at the nominal 5% level when k records carry a corrupted covariate and the model is correct for every other record; the $k = 0$ rows are sizes on the same design. 1000 replicates a scenario; the nominal standard error is 0.007. EDGE is in its unit form at the default partition, with the cubic (default) and symmetric bases, and the Hosmer–Lemeshow statistic is shown at both partitions. The pooled tests combine each record’s residual with those of other records — in groups of predicted risk, or for le Cessie’s test over neighbourhoods in covariate space. Every test built on each record’s own linear predictor rises with k . The last two rows give the median time a data set in milliseconds on one core (the symmetric forms were not timed). Le Cessie’s test is protected but holds an $n \times n$ kernel and grows with the cube of n ; it was run with corrupted records at $n = 1000$ only.

n	k	pooled					record-level			
		EDGE	EDGE-sym	HL (G rule)	HL ($G=10$)	le Cessie	Stukel joint	Stukel sym	GiViTI	cubic LR
1000	0	0.045	0.044	0.038	0.040	0.059	0.044	0.049	0.042	0.041
1000	1	0.056	0.058	0.052	0.051	0.055	0.121	0.115	0.089	0.127
1000	2	0.059	0.067	0.051	0.044	0.052	0.217	0.206	0.146	0.201
1000	5	0.057	0.048	0.045	0.042	0.052	0.398	0.367	0.251	0.321
1000	10	0.097	0.059	0.066	0.058	0.077	0.627	0.592	0.397	0.482
5000	0	0.048	0.053	0.052	0.055	–	0.047	0.052	0.048	0.041
5000	5	0.050	0.045	0.043	0.037	–	0.252	0.222	0.187	0.272
5000	10	0.060	0.052	0.052	0.043	–	0.472	0.420	0.312	0.447
5000	25	0.151	0.099	0.077	0.050	–	0.801	0.784	0.575	0.753
5000	50	0.740	0.745	0.437	0.088	–	0.980	0.977	0.869	0.954
ms, $n = 1000$		4	–	2	2	5,570	1	–	9	5
ms, $n = 5000$		14	–	7	7	1,267,530	2	–	33	19

or dropping the outer groups outright, protects almost completely, between 0.047 and 0.059 at $n = 5000$ with twenty-five and with fifty corrupted records, and costs almost all the power, a mean raw rejection rate of 0.046 to 0.059 over the six link and tail scenarios against the default’s 0.508. Pooling a tail of 1% rather than 5% keeps the power (0.503) and does not protect (0.271 at fifty corrupted records in 5000). The extremes carry the signal, so a remedy that removes them removes the departure with the corruption.

S6 False alarms of every test under the exaggerated covariate

Table S5 gives, for every test of the main comparison, the false-alarm rate under corruption C1 of the main text: k records whose covariate is multiplied by four, in a sample for which the model is otherwise correct. Section 6.1 of the main text reads it.

S7 External validation against the usual tests

When a published model is checked on new patients, the usual tests are Cox’s recalibration test, Spiegelhalter’s z and the GiViTI belt in its external mode. Table S6 compares EDGE in external mode with them, on a design fixed and hashed before it was run. All seven tests hold their level. Over the ten departures EDGE at ten groups has a mean size-adjusted power of 0.415, against 0.420 for the belt, 0.410 for Cox’s test, 0.335 for the Hosmer–Lemeshow statistic, 0.327 for Stukel’s test and 0.271 for Spiegelhalter’s z . The averages hide the shape of the departure. Paired on the same replicates, EDGE at ten groups minus Cox’s test is -0.079 and -0.042 (Monte Carlo standard errors 0.011 and 0.007) on the two shifts, -0.021 and -0.015 on the slopes, -0.068 and -0.141 on the threshold terms, $+0.045$ and $+0.118$ on the U-shaped terms and $+0.053$ and $+0.206$ on the interactions, with standard errors between 0.004 and 0.015; the differences from the belt are within 0.07 of these. The mean difference is $+0.006$ against Cox’s test and -0.005 against the belt, and without the two interactions -0.025 and -0.030 .

The specification wrote three expectations in advance. That Cox’s test, Spiegelhalter’s z and the belt would lead on calibration in the large and the slope held for Cox’s test and the

belt, and for Spiegelhalter’s z only on the slope; on calibration in the large it is last (0.107 and 0.303). That EDGE would be competitive on the U-shaped and threshold terms held for the U-shape and failed for the threshold, where it trails by up to 0.141. That no test would detect the interaction well failed: EDGE has a size-adjusted power of 0.459 against it at the stronger setting at the default partition, and 0.424 at ten groups.

Under corrupted records EDGE at ten groups keeps every rate at or below 0.082, below the bar of 0.10 though above the nominal level. Cox’s test reads 0.088 with five reversed records and 0.158 with ten, the belt 0.192 and 0.347, and Stukel’s test 0.610 with ten. The Hosmer–Lemeshow statistic and Spiegelhalter’s z stay below 0.10 in every setting, with less power. Cox’s test and the belt are not two independent comparators: the belt fits a polynomial of the logit of the prediction whose degree it selects by forward tests at its default threshold of 0.95, and at degree one its model is Cox’s recalibration model, so in about 95% of null replicates, and in 57% to 96% under the alternatives, it returns Cox’s p -value. Its extra false alarms arise when corrupted records push the selected degree above one.

Table S6: External validation: a published model with frozen coefficients checked on new patients. Five standard normal covariates; the truth departs from the published linear predictor in five ways, two strengths each, at $n = 500$; in the last block the model is correct and k of 1000 records carry the second covariate multiplied by four (exaggeration) or by -4 (sign error), with their prediction recomputed at the recorded value. EDGE is in external mode (cubic basis and constant, four degrees of freedom) at ten groups and at the default partition. Cox is the recalibration test on intercept and slope, Spiegelhalter the z test, GiViTI the belt in external mode, HL the Hosmer–Lemeshow statistic at ten groups and Stukel his two terms added to the frozen linear predictor. 1000 replicates a scenario. Power is size-adjusted at the null with $n = 500$; shaded false-alarm rates exceed 0.10.

Scenario	EDGE, $G=10$	EDGE, default	Cox	Spiegelhalter	GiViTI	HL	Stukel
<i>Level: rejection rate of a correct model</i>							
$n = 250$	0.051	0.051	0.053	0.050	0.056	0.048	0.051
$n = 500$	0.051	0.054	0.055	0.057	0.053	0.052	0.062
$n = 1000$	0.061	0.050	0.061	0.061	0.060	0.048	0.052
<i>Power, size-adjusted, $n = 500$</i>							
calibration in the large, $C = 0.2$	0.311	0.298	0.390	0.107	0.392	0.216	0.153
calibration in the large, $C = 0.4$	0.911	0.894	0.953	0.303	0.951	0.770	0.493
calibration slope, $C = 0.8$	0.394	0.387	0.415	0.526	0.415	0.290	0.321
calibration slope, $C = 0.6$	0.972	0.971	0.987	0.994	0.987	0.929	0.956
U-shaped term, $C = 0.4$	0.124	0.127	0.079	0.075	0.084	0.104	0.084
U-shaped term, $C = 0.8$	0.460	0.470	0.342	0.271	0.361	0.335	0.322
threshold term, $C = 2$	0.135	0.120	0.203	0.067	0.202	0.102	0.163
threshold term, $C = 4$	0.277	0.240	0.418	0.086	0.429	0.161	0.282
interaction, $C = 0.6$	0.144	0.151	0.091	0.072	0.102	0.115	0.103
interaction, $C = 1$	0.424	0.459	0.218	0.210	0.279	0.332	0.390
mean of the ten	0.415	0.412	0.410	0.271	0.420	0.335	0.327
<i>False alarms, correct model, $n = 1000$</i>							
exaggeration, $k = 1$	0.044	0.055	0.054	0.048	0.055	0.040	0.048
exaggeration, $k = 2$	0.034	0.043	0.046	0.043	0.047	0.046	0.070
exaggeration, $k = 5$	0.059	0.072	0.050	0.054	0.063	0.048	0.123
exaggeration, $k = 10$	0.055	0.077	0.072	0.070	0.089	0.064	0.173
sign error, $k = 1$	0.045	0.040	0.043	0.047	0.057	0.050	0.114
sign error, $k = 2$	0.044	0.051	0.043	0.044	0.080	0.058	0.132
sign error, $k = 5$	0.065	0.104	0.088	0.071	0.192	0.067	0.349
sign error, $k = 10$	0.082	0.240	0.158	0.090	0.347	0.076	0.610
settings held, of eight	8	6	7	8	6	8	2

S8 EDGE on streaming data

Setting. A frozen model gives each patient a prediction $p_i \in (0, 1)$ before the outcome $y_i \in \{0, 1\}$ is seen. Cut points $0 < c_1 < \dots < c_{G-1} < 1$ are fixed in advance, from reference

predictions or by the analyst, and do not depend on any outcome; patient i falls in group $g(i) = 1 + \#\{j : c_j < p_i\}$. After n patients the monitor holds, for each group,

$$O_g = \sum_{g(i)=g} y_i, \quad E_g = \sum_{g(i)=g} p_i, \quad V_g = \sum_{g(i)=g} p_i(1 - p_i), \quad m_g = \#\{i : g(i) = g\},$$

and computes the external-mode statistic of Section 2 of the main text from them alone: $r_g = (O_g - E_g)/\sqrt{V_g}$ and $\bar{\pi}_g = E_g/m_g$ over the non-empty groups, Z the chosen basis evaluated at $\bar{\pi}_g$ with a constant column, reduced to full column rank k , and $S_n = r^\top P_Z r$.

Proposition. (a) S_n is the same for every order and every division into batches in which the n patients are added, and equals the one-shot external statistic computed on all n patients with the same groups. (b) Suppose that, given all the predictions, past and future, the outcomes are independent with $\Pr(y_i = 1) = p_i$ (the model is calibrated), that G is fixed, and that $V_g \rightarrow \infty$ in every group that is non-empty. Let k_n be the rank of Z , which equals $d + 1$ once $d + 1$ groups with distinct mean risks are non-empty. Then $\sup_x |\Pr(S_n \leq x \mid p_1, \dots, p_n) - F_{\chi_{k_n}^2}(x)| \rightarrow 0$, for any sequence of predictions under which every non-empty group's variance grows, however unequal the groups. (c) Adding a patient costs $O(\log G)$ operations and computing S_n costs $O(Gk^2)$, independently of n .

Proof. (a) Each of O_g, E_g, V_g, m_g is a sum over patients of a quantity that depends on that patient alone, so it does not depend on the order of the terms or on how they are grouped, and S_n is a function of these sums only.

(b) Condition on the predictions, so that the groups, the $\bar{\pi}_g$, the column scaling and rank reduction of Z , and hence Z , are fixed at each n ; this uses that the cut points depend on predictions only. Z still changes with n through the group means, so let Q_n be a $G \times k_n$ matrix whose orthonormal columns span the columns of Z ; then $S_n = \|Q_n^\top r\|^2$ and

$$Q_n^\top r = \sum_{i=1}^n q_{g(i)} \frac{y_i - p_i}{\sqrt{V_{g(i)}}},$$

where q_g is the g th row of Q_n . Given the predictions this is a sum of independent, mean-zero vectors whose covariance is exactly $Q_n^\top Q_n = I_{k_n}$, and each summand is bounded in norm by $\max_g V_g^{-1/2} \rightarrow 0$. For any fixed unit vector a , the scalar sums $a^\top Q_n^\top r$ therefore satisfy Lindeberg's condition, and by the Cramér–Wold device $Q_n^\top r$ is asymptotically $N(0, I_{k_n})$, uniformly over the rotation Q_n because the bound does not involve it. Hence S_n is asymptotically $\chi_{k_n}^2$, in the uniform sense stated. Nothing in the argument uses equal group sizes or a particular distribution of the p_i : every non-empty group need only keep growing. The condition on the outcomes fails if the deployed model changes the care patients receive, or if who arrives depends on earlier outcomes; cut points taken from the predictions of a first batch are allowed, since they use predictions only. Under drift in the risk distribution the conclusion stands, although small groups weaken the protection against corrupted records that equal-frequency groups give (Proposition 1 of the main text bounds a record's influence by the variance of its groupmates).

(c) Finding the group of a new patient is a binary search among $G - 1$ cut points, and updating four sums is constant work. The statistic needs the G residuals, the basis at G points and a QR decomposition of a $G \times k$ matrix. $\square$

Repeated looks. Part (b) concerns one look. The statistics at successive looks share their patients and are dependent, so acting on the first rejection over several looks does not have level α at each look's χ_k^2 critical value; splitting the level, α/L at each of L looks fixed in advance, keeps the overall false-alarm rate at most α in large samples; looks added after the start break this bound. With fixed groups and stable group means the vector r has approximately independent

increments between looks, so group-sequential boundaries for chi-squared statistics (Jennison and Turnbull, 2000, chapter 15) apply and would be less conservative than splitting the level.

Simulation. The specification was fixed and hashed before any replicate was run. Reference predictions were 10,000 draws with $\text{logit } p \sim N(-1.5, 1)$, and their deciles gave the cut points; patients then arrived in batches of 100, 2000 replicates a cell. Under a calibrated model the level at 5% was 0.047 after 10,000 patients and 0.054 after 100,000; when every monitored patient had $\text{logit } p \sim N(-0.5, 1.3)$, a distribution different from the reference, so that the smallest group held about 3900 of 100,000, it was 0.050 and 0.052. Testing after each of ten blocks of 1000 patients and stopping at the first rejection gave a false-alarm rate of 0.214 at 5% a look and 0.030 at 0.5% a look; the last look alone read 0.057. Against an overfitted model, $\text{logit } \Pr(y = 1) = 0.8 \text{logit } p$, the test rejected in 0.993 of replicates after 2000 patients and in all after 10,000. The median time to add a hundred patients was about 0.5 milliseconds and to compute the test about 1.5 milliseconds, on one core of an AMD Ryzen 9 3900X; both are independent of the number of patients already added, which was a million when they were timed.

S9 Power by departure family

Table S7 replaces the counts of scenarios in which EDGE leads or ties a comparator by the mean difference in size-adjusted power over each departure family, with a Monte Carlo interval. The counts in Figures 1 and 2 of the main text use a margin of 0.01, which is close to the Monte Carlo standard error of a single difference, so they do not separate a small loss from a tie; the means do. The last column is what ten groups cost against the default partition on clean data.

Table S7: Mean difference in size-adjusted power, EDGE (cubic basis, unit form, default partition) minus each comparator, by departure family, with a Monte Carlo 95% interval in brackets. The interval ignores the positive correlation between two tests read on the same replicates, which makes it conservative, and the Monte Carlo error of the critical values used for size adjustment. The last column is the price of ten groups.

Family	Scenarios	HL, ten groups	Stukel joint	Stukel one-par.	cubic LR	GiViTi	EDGE, ten groups
symmetric tails	32	+0.175 (0.171, 0.179)	-0.007 (-0.011, -0.003)	-0.062 (-0.066, -0.058)	+0.001 (-0.003, 0.005)	+0.105 (0.101, 0.109)	+0.036 (0.032, 0.040)
asymmetric links	21	+0.183 (0.180, 0.187)	-0.010 (-0.014, -0.006)	+0.183 (0.179, 0.187)	-0.072 (-0.076, -0.068)	-0.075 (-0.079, -0.071)	+0.024 (0.020, 0.028)
Stukel symmetric family	10	+0.127 (0.123, 0.131)	-0.002 (-0.006, 0.002)	-0.054 (-0.058, -0.050)	-0.022 (-0.026, -0.018)	+0.107 (0.103, 0.111)	+0.019 (0.015, 0.023)
omitted terms	80	+0.067 (0.066, 0.069)	-0.015 (-0.017, -0.014)	-0.026 (-0.027, -0.025)	+0.010 (0.008, 0.011)	-0.022 (-0.023, -0.020)	+0.011 (0.009, 0.012)
rough misfit	13	-0.461 (-0.464, -0.458)	-0.008 (-0.012, -0.004)	-0.038 (-0.042, -0.034)	-0.023 (-0.027, -0.020)	+0.093 (0.089, 0.097)	-0.000 (-0.004, 0.003)
off-index	2	+0.026 (0.019, 0.034)	-0.002 (-0.010, 0.006)	+0.031 (0.024, 0.038)	+0.001 (-0.007, 0.009)	-0.007 (-0.016, 0.001)	+0.018 (0.010, 0.025)
all families	158	+0.064 (0.063, 0.066)	-0.011 (-0.013, -0.010)	-0.007 (-0.009, -0.006)	-0.008 (-0.009, -0.007)	+0.015 (0.014, 0.016)	+0.017 (0.016, 0.018)

S10 What protects a test

Design. The study was specified and hashed before any replicate ran, with two addenda written before their replicates: one adding a milder diagnostics rule after a smoke run of three replicates showed that the declared rule removes about 3.5% of a clean data set, the other adding the tests of Hosmer and Hjort. Every test runs on the same data sets, 1000 replicates a cell, so all comparisons are paired. The two-covariate design is that of the corruption study of the main text; the twelve-covariate design has eight standard normal and four Bernoulli(0.3) covariates at $n = 2000$, with the error applied to the first covariate. The ungrouped twin is the score test for adding the orthogonal cubic polynomial in the fitted risk to the model, on three degrees of freedom. The tests of Hosmer and Hjort use their general-purpose weight $\hat{\pi} \log \hat{\pi}$ at ten groups, in their Hosmer–Lemeshow form on $G - 2$ degrees of freedom and their full-covariance form; their implementation, written from the paper, held its level within 0.05 ± 0.02 at $n = 1000$ and 5000 before the study.

Results. Table S8 gives the rates. The grouped tests at ten groups, EDGE in its unit form and the Hosmer–Lemeshow statistic, stay below 0.10 in every setting with exaggerated covariates. Under sign errors EDGE at ten groups holds with one reversed record and with ten among twelve covariates, and reads 0.210 with ten reversed records in 1000 and 0.236 with twenty-five in 5000; the Hosmer–Lemeshow statistic reads 0.093 and 0.120 there. Every ungrouped test, whatever the scale of its directions, fails at one or two corrupted records once the model is refitted. On frozen predictions (Table S9) the ungrouped twin holds against exaggerated covariates but not against five or more reversed records, and Spiegelhalter’s z holds throughout. This frozen-prediction arm was added by a third addendum, hashed before it ran, on the data sets of the external-validation study (S7), which it regenerates and checks against that study’s Spiegelhalter p -values. Spiegelhalter’s z rejects in none of the 25,000 replicates at its nominal level, because its normal reference ignores the fitted coefficients; at its own realised level it is as fragile as Stukel’s test. The tests of Hosmer and Hjort group the records but weight each within its group by a function of its covariates, and fail at about five reversed records in 1000. EDGE’s p -values from the Satterthwaite and the Imhof reference gave identical rejections at 5% in every cell.

Influence diagnostics. Removing every record with Pregibon’s $\Delta X^2 > 4$ or $\Delta\beta > 1$, the declared rule, removed 3.7% of the records of a clean data set at $n = 1000$ and every test then rejected a correct model in more than 0.95 of replicates, because the records removed are those whose outcome is least likely under the model. Removing only the records with $\Delta\beta > 1$, the addendum’s rule, removed none in any cell, so it changed nothing.

The Pulkstenis–Robinson test. It was run in every scenario of the main battery and is reported here. Its mean size-adjusted power over the 158 alternatives is 0.16, against 0.54 for EDGE at the default partition and 0.48 for the Hosmer–Lemeshow statistic at ten groups. It groups on covariate patterns (Pulkstenis and Robinson, 2002); against exaggerated covariates it stays below 0.10 up to ten records in 1000 and twenty-five in 5000 and reaches 0.121 at fifty in 5000, and it is not protected against sign errors: 0.224 with ten reversed records in 1000 and 0.888 with fifty in 5000.

Table S8: What protects a test. Rejection rate at 5% on the same data sets, 1000 replicates a row: false alarms under a correct model, with k records carrying the covariate multiplied by 4 or -4 (shaded above 0.10), and size-adjusted power under misfit (last six rows). EDGE at the default partition and at ten groups; HL, the Hosmer–Lemeshow statistic at ten groups; HH, the weighted grouped tests of Hosmer and Hjort with their general-purpose weight; Twin, the ungrouped score test for the cubic directions of EDGE in the predicted risk; EDGE $_n$, EDGE with one record a group ($n \leq 1000$); Spz, Spiegelhalter’s z , whose raw rate is 0 throughout and which is shown at its own realised level.

Design	n	Setting	EDGE	EDGE $_{10}$	HL	HH $_{HL}$	HH $_{X^2}$	Twin	EDGE $_n$	Spz	Stukel	cubic	GiViTI
two covariates	1000	correct model	0.063	0.059	0.052	0.055	0.055	0.057	0.057	0.050	0.063	0.058	0.049
two covariates	1000	$\times 4, k = 1$	0.047	0.051	0.042	0.052	0.059	0.166	0.148	0.118	0.147	0.149	0.111
two covariates	1000	$\times 4, k = 10$	0.101	0.066	0.049	0.183	0.271	0.641	0.594	0.546	0.595	0.454	0.375
two covariates	1000	$\times -4, k = 1$	0.051	0.041	0.043	0.073	0.141	0.468	0.379	0.303	0.378	0.393	0.278
two covariates	1000	$\times -4, k = 10$	0.406	0.210	0.093	0.676	0.815	0.992	0.986	0.987	0.986	0.978	0.966
two covariates	5000	correct model	0.049	0.050	0.027	0.037	0.038	0.052	–	0.050	0.053	0.048	0.051
two covariates	5000	$\times 4, k = 50$	0.745	0.096	0.069	0.456	0.615	0.989	–	0.966	0.973	0.953	0.879
two covariates	5000	$\times -4, k = 25$	0.777	0.236	0.120	0.819	0.946	1.000	–	1.000	1.000	1.000	1.000
twelve covariates	2000	correct model	0.045	0.065	0.050	0.065	0.070	0.057	–	0.050	0.043	0.046	0.042
twelve covariates	2000	$\times 4, k = 25$	0.102	0.055	0.054	0.061	0.072	0.218	–	0.108	0.275	0.217	0.134
twelve covariates	2000	$\times -4, k = 10$	0.091	0.048	0.058	0.075	0.081	0.292	–	0.134	0.315	0.306	0.204
two covariates	1000	cloglog truth	0.738	0.708	0.382	0.286	0.233	0.684	0.756	0.987	0.765	0.837	0.874
two covariates	1000	top 2%, risk 0.6 of prediction	0.416	0.194	0.166	0.224	0.209	0.742	0.469	0.380	0.432	0.509	0.422
two covariates	1000	C2, $k = 10$	0.601	0.261	0.222	0.334	0.323	0.988	0.727	0.586	0.673	0.813	0.683
two covariates	5000	top 2%, risk 0.8 of prediction	0.624	0.292	0.315	0.364	0.372	0.886	–	0.661	0.578	0.718	0.526
twelve covariates	2000	cloglog truth	0.552	0.379	0.232	0.256	0.241	0.425	–	0.225	0.545	0.622	0.614
twelve covariates	2000	omitted interaction	0.464	0.276	0.155	0.209	0.189	0.364	–	0.260	0.490	0.483	0.587

Table S9: The same question on frozen predictions: rejection rate at 5%, 1000 replicates a row, on the corrupted data sets of the external-validation study ($n = 1000$, five covariates), where nothing is refitted. Twin, the score test for the cubic directions of EDGE in the frozen prediction, on three degrees of freedom; Spz, Spiegelhalter’s z ; EDGE in external mode at ten groups and at the default partition. Shaded above 0.10.

Setting	Twin	Spz	EDGE ₁₀	EDGE
correct model	0.052	0.061	0.061	0.050
$\times 4, k = 1$	0.061	0.048	0.044	0.055
$\times 4, k = 2$	0.044	0.043	0.034	0.043
$\times 4, k = 5$	0.067	0.054	0.059	0.072
$\times 4, k = 10$	0.096	0.070	0.055	0.077
$\times - 4, k = 1$	0.051	0.047	0.045	0.040
$\times - 4, k = 2$	0.057	0.044	0.044	0.051
$\times - 4, k = 5$	0.132	0.071	0.065	0.104
$\times - 4, k = 10$	0.311	0.090	0.082	0.240

S11 Proof of Theorem 2

Notation and assumptions are those of Appendix A of the main text: φ are the shape functions of Theorem 2, $M = \int_0^1 \varphi \varphi^\top dq$, $K = \int_0^1 \varphi \kappa^\top dq$, and the limit to be shown is (A1) of the main text.

Step 0: the bases Because P_Z depends on Z only through its column space, and (Step 6) so do the weights, the columns may be taken as $Z_{gj} = \varphi_j(q_g)$ once the bases of Section 2.2 of the main text are shown to span that space up to a uniform $o(1)$. They evaluate their shapes at $\bar{\pi}_g$ or $\bar{\eta}_g = \text{logit } \bar{\pi}_g$ rather than at q_g . Under (A3) and (A4′) the empirical quantile function of the fitted risk satisfies $\sup_q |\hat{Q}_{\hat{\pi}}(q) - Q_\pi(q)| = O_p(n^{-1/2})$ and Q_π is Lipschitz, so averaging over a group of width $1/G$ gives $\bar{\pi}_g = Q_\pi(q_g) + O(1/G) + O_p(n^{-1/2})$ uniformly in g ; with shape functions continuously differentiable this gives $z_{gj} = \varphi_j(q_g) + \delta_{gj}$ with $\sup_g \|\delta_g\| = O(1/G) + O_p(n^{-1/2})$, where φ_j is the shape composed with Q_π (or with Q_η for the Stukel columns). The contribution of δ to ζ_n below is a sum whose summands are bounded by $Cn^{-1/2} \sup_g \|\delta_g\|$, handled as replacement (a) of Step 4, and its contribution to $G^{-1}Z^\top Z$ and to Step 5 is $O(1/G) + O_p(n^{-1/2})$ by the same bound. Orthonormalising the columns on the observed $\{\bar{\pi}_g\}$ multiplies Z by an invertible matrix that converges at the same rate, and S and the weights are invariant to it. For the Stukel basis linear independence needs 0 interior to the support of η ; otherwise one sign-split column vanishes identically and is dropped.

Step 1: the statistic as a smooth-weighted sum Put $\zeta_n = G^{-1/2}Z^\top r$, so that $S = \zeta_n^\top (G^{-1}Z^\top Z)^{-1} \zeta_n$ with $G^{-1}Z^\top Z \rightarrow M$, a Riemann sum, and M invertible because the φ_j are linearly independent. By Step 1 of the proof of Theorem 1 of the main text, $r = \tilde{r} - U(X^\top W X)^{-1} s + \rho$, hence

$$\zeta_n = \sum_{i=1}^n w_i \varepsilon_i - [G^{-1/2}Z^\top U(X^\top W X)^{-1/2}] [(X^\top W X)^{-1/2} s] + G^{-1/2}Z^\top \rho, \quad w_i = \frac{G^{-1/2} z_{g(i)}}{\sqrt{V_{g(i)}}} \in \mathbb{R}^d. \quad (1)$$

Under (A3) the true linear predictor has compact support and $v_i \geq c_0 > 0$ for every i . Since $\mathbb{E}\|x\|^2 < \infty$ gives $\max_{i \leq n} \|x_i\| = o_p(\sqrt{n})$, and expit is $\frac{1}{4}$ -Lipschitz, $\sup_i |\hat{\pi}_i - \pi_i| \leq \frac{1}{4} \max_i \|x_i\| \|\hat{\beta} - \beta\| = o_p(1)$, so $\sup_i |\hat{v}_i/v_i - 1| = o_p(1)$. On an event of probability tending to one, therefore, $V_g \geq \frac{1}{2}c_0|g|$ simultaneously in g , whatever the group sizes and including $|g| = 1$, and the normalisation computed from $\hat{\pi}$ differs from the same sum with π by a factor $1 + o_p(1)$

uniformly in g . Since φ is bounded and $|g| = n/G$, $\|w_i\| \leq C G^{-1/2}(n/G)^{-1/2} = C n^{-1/2}$ for every i , whatever the group size.

Step 2: the remainder The second-order term of the expansion of $\hat{\pi}_i$ gives $|\rho_g| \leq C V_g^{-1/2} \sum_{i \in g} \|x_i\|^2 \|\hat{\beta} - \beta\|^2$, so $\|G^{-1/2} Z^\top \rho\| \leq C n^{-1/2} \|\hat{\beta} - \beta\|^2 \sum_i \|x_i\|^2 = O_p(n^{-1/2})$. Only $\sum_i \|x_i\|^2 = O_p(n)$, which (A2) supplies, is used; no bound within a single group is needed. The $o_p(n^{-1/2})$ of the expansion in (A2) contributes $\|G^{-1/2} Z^\top U(X^\top W X)^{-1/2}\| O_p(\sqrt{n}) o_p(n^{-1/2}) = o_p(1)$ by Step 5, and replacing v_i by the $\hat{v}_i$ carried in U contributes at most $\sum_i \|w_i\| |\hat{v}_i - v_i| \|x_i\| \|\hat{\beta} - \beta\| \leq C n^{-1/2} o_p(1) O_p(n) O_p(n^{-1/2}) = o_p(1)$ by Step 1.

Step 3: the central limit theorem with population weights Let $w_i^0 = n^{-1/2} \varphi(q_i) / \sqrt{v_i}$ and $\xi_n^0 = \sum_i w_i^0 \varepsilon_i$. The pairs $(w_i^0 \varepsilon_i, n^{-1/2} J_0^{-1/2} x_i \varepsilon_i)$ are independent and identically distributed, w_i^0 being a fixed function of x_i through q_i , with finite variance by $\mathbb{E}\|x\|^2 < \infty$, so the Lindeberg–Lévy theorem and the Cramér–Wold device give joint asymptotic normality. Ranking by $\hat{\pi}$ is ranking by $\hat{\eta}$, and $F_\pi(\pi_i) = F_\eta(\eta_i)$, so the quantile position of the predicted risk is q_i ; by (A4'), $v_i = v(q_i)$. By the law of large numbers $\text{Var}(\xi_n^0) \rightarrow \mathbb{E}[\varphi(q)\varphi(q)^\top] = M$, q being uniform, and $\text{Cov}(\xi_n^0, n^{-1/2} J_0^{-1/2} s) \rightarrow \mathbb{E}[\varphi(q)\sqrt{v(q)}x^\top] J_0^{-1/2} = K J_0^{-1/2}$. Hence $(\xi_n^0, n^{-1/2} J_0^{-1/2} s) \rightsquigarrow N(0, \begin{pmatrix} M & K J_0^{-1/2} \\ J_0^{-1/2} K^\top & I \end{pmatrix})$.

Step 4: from population weights to the weights of the statistic Three replacements separate $\sum_i w_i \varepsilon_i$ from ξ_n^0 : the position q_i is replaced by the rank fraction R_i/n ; the rank fraction by its group position $q_{g(i)}$; and the ranks and the normalisation are computed under $\hat{\beta}$ rather than β . (a) Since $|\varphi(R_i/n) - \varphi(q_{g(i)})| \leq C/G$, at a fixed b the second replacement changes the sum by a quantity with conditional mean zero and conditional variance at most $C/G^2 \rightarrow 0$; its version uniform in b is part of (c). (b) At fixed β the ranks are functions of X alone, and $\Delta_i = \varphi(R_i/n) - \varphi(q_i)$ satisfies $|\Delta_i| \leq C \sup_t |\hat{F}_n(t) - F_\eta(t)| = O_p(n^{-1/2})$ by the Dvoretzky–Kiefer–Wolfowitz inequality, so the conditional variance of $\sum_i n^{-1/2} \Delta_i \varepsilon_i / \sqrt{v_i}$ is $n^{-1} \sum_i \Delta_i^2 = O_p(n^{-1})$. (c) These bounds hold at a fixed b , where the ranks are X -measurable; the statistic uses $\hat{\beta}$, so uniformity over $\{\|b - \beta\| \leq C n^{-1/2}\}$ is needed, and $\hat{\beta}$ lies in that ball with probability tending to one by (A2). Write $T_n(b) = \sum_i \{w_i(b) - w_i^0\} \varepsilon_i$. The weight $w_i(b)$ depends on b through the group index $g_b(i)$ and through $\hat{\pi}_i(b)$. The group indices depend on b only through the ordering of $\{x_i^\top b\}$, and the number of distinct orderings is the number of cells of the arrangement of the $\binom{n}{2}$ hyperplanes $\{b : x_i^\top b = x_j^\top b\}$ in $\mathbb{R}^p$, at most $\sum_{j \leq p} \binom{n}{j} = O(n^{2p})$ (Cover, 1965): polynomially many, so a union bound replaces any chaining argument. Within one cell, fixed at a point b_0 , the group indices are constant and $b \mapsto w_i(b)$ is C^1 through $\hat{\pi}_i(b)$, with first derivative bounded by $C n^{-1/2} \|x_i\|$ and second by $C n^{-1/2} \|x_i\|^2$, so $T_n(b) - T_n(b_0)$ is a linear term in $b - b_0$ plus $O_p(n^{-1/2})$. The coefficient of the linear term is $\sum_i \{\partial w_i(b_0) / \partial b\} \varepsilon_i$, whose weights are X -measurable within the cell and bounded by $C n^{-1/2} \|x_i\|$, so the union bound below applies to it as well, uniformly over cells; multiplied by $\|b - b_0\| \leq C n^{-1/2}$ it is $O_p(\sqrt{\log n} n^{-1/2})$. At b_0 both the linear term's coefficient and $T_n(b_0)$ are sums of X -measurable weights times the centred ε_i , with $\sum_i \{w_i(b_0) - w_i^0\}^2 = O_p(G_n^{-2} + n^{-1})$. The two terms come from the two replacements: the group position differs from the rank fraction by at most $1/(2G_n)$, which by (a) contributes $O(G_n^{-2})$, and the rank fraction differs from the population position by at most $\sup_t |\hat{F}_{n,b}(t) - F_b(t)| + C \|x_i\| \|b - \beta\|$, where the first term is $O_p(n^{-1/2})$ uniformly in b because the half-spaces $\{x^\top b \leq t\}$ form a Vapnik–Červonenkis class (van der Vaart and Wellner, 1996, Section 2.6). The second term is $o_p(1)$ uniformly in i under the moment condition of (A2) but not $O_p(n^{-1/2})$ uniformly; the bound needs only its sum of squares, $n^{-1} \sum_i \|x_i\|^2 = O_p(1)$, which (A2) gives. Hoeffding's inequality conditionally on X gives sub-Gaussian tails for both at each b_0 , and a union bound over the $O(n^{2p})$ cells gives $\sup_{\|b - \beta\| \leq C n^{-1/2}} |T_n(b)| = O_p\{\sqrt{\log n} (G_n^{-1} + n^{-1/2})\}$,

which is $o_p(1)$ under the rate condition $G_n/\sqrt{\log n} \rightarrow \infty$ of Theorem 2 of the main text; the same bound covers the normalisation V_g under $\hat{\beta}$. The group indicators do not form a class of fixed complexity as $G \rightarrow \infty$, but the ordering they induce does, which is the random-cell problem of Moore and Spruill (1975) in a regime their fixed-cell argument does not cover. Together, $\sum_i w_i \varepsilon_i = \xi_n^0 + o_p(1)$.

Step 5: the fitting correction The g -th summand of $Z^\top U$ is $z_g V_g^{-1/2} \sum_{i \in g} \hat{v}_i x_i^\top$, so $G^{-1/2} Z^\top U (X^\top W X)^{-1/2} = \sum_i w_i \hat{v}_i x_i^\top (X^\top W X)^{-1/2}$, a sum over all n observations. Write $\psi = \varphi/\sqrt{v}$, uniformly continuous on $[0, 1]$. Then $w_i = n^{-1/2} \psi(q_{g(i)}) \{1 + o_p(1)\}$ uniformly in i , because $\{V_g/|g|\}^{-1/2} = v(q_g)^{-1/2} + o_p(1)$ uniformly in g : every observation in g has a position within $1/G$ of q_g , $v \circ Q_\eta$ is uniformly continuous by (A3) and (A4'), and $\sup_i |\hat{v}_i - v_i| = o_p(1)$. Likewise $|q_{g(i)} - q_i| \leq 1/G + \sup_t |\hat{F}_n(t) - F_\eta(t)| + C \sup_i |\hat{\eta}_i - \eta_i| = o_p(1)$ uniformly in i , so $\delta_n = \sup_i |\psi(q_{g(i)}) - \psi(q_i)| = o_p(1)$ and

$$\left\| \sum_i w_i \hat{v}_i x_i^\top - n^{-1} \sum_i \varphi(q_i) \sqrt{v_i} x_i^\top \right\| \leq \{\delta_n + o_p(1)\} n^{-1} \sum_i \|x_i\| = o_p(1).$$

Since $n^{-1} \sum_i \varphi(q_i) \sqrt{v_i} x_i^\top \rightarrow K$ and $(X^\top W X)^{-1/2} = n^{-1/2} J_0^{-1/2} \{1 + o_p(1)\}$, the bracket in (1) converges to $K J_0^{-1/2}$. The law of large numbers used is across the whole sample and not within a group, which is why the number of observations per group need not grow.

Step 6: conclusion From (1) and Steps 2–5, $\zeta_n = \xi_n^0 - K J_0^{-1/2} (n^{-1/2} J_0^{-1/2} s) + o_p(1) \rightsquigarrow N(0, \Sigma)$, and by the continuous-mapping theorem $S \rightsquigarrow \zeta^\top M^{-1} \zeta$ with $\zeta \sim N(0, \Sigma)$, which is $\sum_j \lambda_j \chi_{1,j}^2$ with λ_j the eigenvalues of $M^{-1} \Sigma$. These do not depend on the basis chosen for $\text{col}(Z)$: for invertible A , $\{(ZA)^\top Z A\}^{-1} (ZA)^\top \Omega Z A = A^{-1} \{(Z^\top Z)^{-1} Z^\top \Omega Z\} A$ is a similarity transformation. The plug-in weights at G_n , the eigenvalues of $(G^{-1} Z^\top Z)^{-1} (G^{-1} Z^\top \hat{\Omega} Z)$, converge to the λ_j : $G^{-1} Z^\top Z \rightarrow M$ as a midpoint Riemann sum, and $G^{-1} Z^\top \hat{\Omega} Z \rightarrow M - K J_0^{-1} K^\top$ by Step 5. The bounds follow: $K J_0^{-1} K^\top \succeq 0$ gives $\Sigma \preceq M$ and hence $\lambda_j \leq 1$, while $\Sigma \succeq 0$ as the covariance of the limit of ζ_n . A weight is zero exactly when $c^\top \varphi(q) = \sqrt{v(q)} \{a_0 + a_1 Q_\eta(q)\}$ for some a_0, a_1 , the population form of the level shift and tilt the fit absorbs; a zero weight removes a degree of freedom from the limit and changes neither $\sum_j \lambda_j$ nor $\sum_j \lambda_j^2$, so the Satterthwaite and Imhof evaluations are unaffected. $\square$

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